# Wealth Taxation & Labor Supply: Evidence from the Danish Wealth Tax Experience

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**Abstract**

I estimate the effect of wealth taxation on labor supply incentives, focusing on the efficiency argument advanced in the New Dynamic Public Finance literature. This argument to tax wealth holds that, under private information and stochastic earnings ability, an individual's optimal savings decision reduces their future labor supply incentives due to the absence of an insurance market, leading to an inefficiency. Using a difference-in-difference strategy that doubled the threshold for wealth tax liability for couples in Denmark in 1989, I find a null labor income response, . . .

Using GMM estimator, I estimate consumption Euler equations to rule between a lifecycle model with an endogenously missing market due to moral hazard and a full insurance benchmark. I find that while the theoretical mechanism is plausible, its empirical relevance for Denmark, and potentially other developed economies, is limited. Using a difference-in-difference strategy, I find a null

*JEL: .*

*Keywords: Wealth Taxation; Labor Supply; Dynamic Public Finance; Inverse Euler Equation.*

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# 1 Introduction

Recent years have seen a surge in political and academic interest around wealth taxation, spurred by growing economic inequality and calls for policy tools to address wealth concentration. While much of this debate has focussed on redistributive goals, a gap remains in our understanding of the broader economic impacts of wealth taxation – particularly, its impact on labor supply decisions (Scheuer & Slemrod 2021). This paper addresses a key question in public finance: how do wealth taxes influence household labor supply incentives, including those of the primary and secondary household earners, as well as decisions such as retirement and self-employment? Understanding this relationship is crucial for designing tax policies that balance equity and efficiency, especially because multiple theories of economic behavior defend a Pareto-improving wealth tax to incentivise labor decisions (Golosov et al. 2006).

I find

This is particularly relevant in the context of New Dynamic Public Finance models, which argue that wealth taxation can counteract inefficiencies arising from private information and stochastic earning abilities. While few empirical papers has primarily focused on wealth accumulation in response to taxation (Brülhart et al. 2019, Ring 2020, Jakobsen et al. 2020), few studies have examined labor supply effects. This paper is the first to empirically test this dynamic using administrative data from Denmark. Two Democratic political candidates in the US primaries indicated their support for wealth tax proposals. A majority of Americans currently favor a wealth tax on the very rich. Academically, there is a growing literature documenting the rise of wealth inequality (Saez & Zucman 2020) and studying the impact of wealth tax changes (Jakobsen et al. 2020). This literature focuses on the dynamics of wealth accumulation and its response to wealth taxation. However, little work studied the efficiency implications of taxing wealth. In this paper, I study the labor supply response to wealth taxation. I provide evidence on a common efficiency argument to tax wealth. Research work on understanding the different mechanisms of adjustment in response to wealth tax changes is vital for the design of policy on wealth taxes (Adam & Miller 2020).

**[PARAGRAPH ABOUT RESULTS.]**

A classic argument to tax wealth in standard dynamic economic models relies on inducing incentives to work (Farhi & Werning 2013*b*). In those private information economies, left to their own devices individuals would over-save in order to hedge against the uncertainty of their future earnings profile. This then reduces their labor supply incentives on the resolution of uncertainty. Wealth taxation thus acts to dis-incentivise Pareto-dominated over-saving. This argument to tax wealth in the "New Dynamic Public Finance" literature rests on efficiency concerns (Rogerson 1985, Golosov et al. 2003, Farhi & Werning 2013*b*). Assuming private information and the stochastic evolution of an individual's ability to generate income drives individuals to over-save for insurance purposes. Two key assumptions lead to this result: the stochastic evolution of ability; and the lack of a private insurance market to hedge against the risk of a downfall in future ability to generate income. Using Danish Administrative data, this paper tests this Dynamic Mirreleesian lifecycle model with uncertainty and private information against a full information benchmark, where the wealth tax result reverts to zero in equilibrium (Mirrlees 1971, Saez 2001, Golosov et al. 2003).

I provide evidence on the effect of wealth taxation on labor supply incentives at different margins of response including household earnings, retirement, and self-employment decisions. I then investigate the implications of this evidence for multiple competing economic theories of behavior in response to wealth taxes. In particular, I answer the question of whether an endogenously incomplete markets economy due to a moral hazard inefficiency fits the data better than a complete markets economy. Thus, I provide direct evidence on one important argument to tax wealth, aiding policymakers in designing policy and academics in assessing competing theories of economic behavior.

**[PARAGRAPH ABOUT DENMARK WEALTH TAX CHANGE AND ID.]**
**[PARAGRAPH ABOUT METHOD FOR MORAL HAZARD VS. FULL INFO.]**

# 2 Literature Background

**Reduced-form work on the effect of wealth and labor taxation on labor income**
Work on the effect of wealth taxes on labor supply outcomes is nascent and doesn't address the efficiency implications of taxing wealth. However, a recent literature that focusses on the impact of wealth taxes on individual behavior emerged, owing to increasingly available high quality data on the joint distributions of income, consumption, and wealth. Brülhart et al. (2019) study a reduction in the wealth tax in one Swiss canton relative to another. They find no labor earnings response to a reduction in wealth taxes. In their study, wealth is self-reported to the tax authority leaving a lot of room for evasion concerns. Ring (2020) studies a change in the tax assessment of housing wealth in Norway that differentially affected household's estimated wealth. He finds that an increase in exposure to wealth taxes led to increased labor earnings. One concern with this procedure of identifying the effect is that housing provides a consumption good. Hence, exposure to a tax on housing wealth might not alter the individual's intertemporal decisions. In both Brülhart et al. (2019) and Ring (2020), an additional concern with the estimates is the identification argument relies on location decisions and assumes their rigidity. Seim (2017) also investigates the impact of a wealth tax reform using Swedish Administrative data. His focus is on the savings behavior and evasion responses.

Furthermore, an established literature focuses on the labor supply responses to labor income taxes (see Keane (2011) for a summary). Consensus estimates indicate that men supply labor inelastically and women have a higher elasticity in response to taxation. Also, there is evidence that extensive margin responses tend to be more meaningful than intensive margin responses.

The Danish wealth tax reform informs the literature on the impact of wealth taxation on labor supply in multiple ways. Firstly, the Danish wealth tax reforms were not tied to locations. Hence, location choices within Denmark would not be a concern for identification. Secondly, the third party reporting of most assets reduces the evasion responses of individuals. Thirdly, the availability of data allows me to study responses to the tax cut at different margins of labor supply like spousal labor supply, self-employment, and early retirement decisions. Although one concern is that the wealth tax was paid only by the top 2-3% of the distribution, reducing scope for a labor supply response, it is likely that a reduction in wealth tax (like the Danish experiment) will reduce the incentives to work for those adults most, since their wealth can generate enough returns for them to not need labor income. Additionally, this paper would be the first to only consider the labor supply response to wealth taxation uniquely as opposed to papers that focus more on savings behavior and wealth accumulation. (Jakobsen et al. 2020, Seim 2017)

Despite a large literature in the macroeconomics/public finance theory fields, past empirical literature comparing an endogenously incomplete market informational inefficiency to a self insurance benchmark is little (Attanasio & Weber 2010). In fact, empirical literature assessing this as an argument to tax wealth is non-existent as far as I am aware. This probably owes to two reasons. Firstly, data on wealth is extremely rare and often unreliable. Secondly, until recently, computational power put severe constraints on assessing these competing theories. Multiple papers however attempted to deal with the technical challenges and tested models of endogenously incomplete markets against those of complete markets. However, those were mainly conducted in developing country contexts.

Ligon (1998), concerned with risk-sharing in three village economies in India studied by

Townsend (1994), uses data on consumption to answer whether the standard Euler equation or the inverse Euler equation applies. In doing so, Ligon (1998) tests between three different models (Private Information, Permanent Income, and Full Insurance) using GMM, imposing restrictions on the consumption behavior implied by the alternative models. His findings favor a private information world for two of the villages and permanent income hypothesis for the final one. The study is naturally not concerned with labor supply decisions nor wealth taxation. It aims to quantify the extent to which social/kinship-based insurance is substituting for government in a developing country context. A further body of work in that tradition followed suit. Kinnan (2021) studies consumption and risk-sharing in similar context in Thailand. Surprisingly, in those studies, there is no interest in the interaction of these effect with labor supply incentives and decisions.

A wider empirical literature concerned with consumption and risk-sharing exists (See Attanasio & Weber (2010) for a summary). In the UK, Attanasio & Pavoni (2011) explain the excess smoothness of consumption (Campbell & Deaton 1989), using models of moral hazard similar to the dynamic Mirreleesian model. Using US data, Blundell et al. (2008) find evidence of some partial insurance against permanent income shocks. However, as Attanasio & Weber (2010) conclude research on models with asymmetric information and their implications "is only beginning." Empirical studies that attempt to use wealth tax changes to answer questions within the framework of these models are entirely non-existent. This is while the dynamic public finance agenda has achieved significant progress that yields testable predictions and have implications for capital taxation (Golosov & Tsyvinski 2015).

Methodologically, Karaivanov & Townsend (2014) provide and apply a method to compare across multiple informational regimes and financial structures. Based on maximum likelihood and linear programming, their approach is categorised by optimising the model subject to the original incentive compatibility constraints rather than using the common first-order approach (used by Rogerson (1985) and Ligon (1998) in public finance applications). These lead to an environment that allows for measurement error, unobserved heterogeneity, and non-convexities in the data. Another attractive feature of their method is its ex ante agnostic about the underlying data generation process. Although the application focuses mostly on financial constraints and their implications for insurance in developing Thailand. The method can be employed in the current exercise. The costs of this methodology is the curse of dimensionality that immediately bites as grids of state variables become more fine. The added benefit that this paper can bring-forth is that there are wealth tax changes that can imply 'clean' behavioral labor supply responses, coupled with an assessment of the inverse Euler equation argument to tax wealth.

More generally, this paper is related to a body of literature that assesses the impact of capital taxation on economic distortions. **Discuss New Dynamic Public Finance**

# 3 Institutional Setting

## 3.1 Danish Wealth Tax Reforms

I employ three Identification strategies to study the effect of wealth tax changes on labor supply incentives using reforms that occurred in the 1980s and 1990s in Denmark. I detail the wealth tax changes and how they can be potentially used to distinguish between the different informational regimes.

- Change 1: relies on a 1989 reform that doubled the threshold for which couples start being liable for paying a wealth tax. This was done to be more fair for couples. Therefore couples within a threshold suddenly had their 2.2% wealth tax reduced to zero.

- Change 2: the tax on wealth was also reduced over 3 years between 1989-1991 from 2.2% to 1%.

- Change 3: the wealth tax was abolished in 1997.

Potential ID strategies:

- ID1: Using the first policy change, define the treated households with in the old wealth threshold but outside the new wealth threshold. Define the control group as singles in the same wealth threshold who continued to pay the tax. Treatment period is 1989.
- ID2: Using the first policy change, define the treated households with in the old wealth threshold but outside the new wealth threshold. Define the control group as couple just below the original wealth threshold. Treatment period is 1989.
- ID3: Denmark operates a horizontal upper tax cap. Use households above the horizontal upper cap as part of a control group around 1989. This is because their marginal wealth tax is zero. Individuals just below the threshold pay a wealth tax of 2.2% before 1989. This was then reduced to 1% through to 1991.
- ID4: Apply ID3 to the abolishment of the wealth tax in 1997 in Denmark.

ID1-3 were adopted previously by Jakobsen et al. (2020) to study the effect of wealth tax on wealth accumulation. ID4 was not used previously since the abolishment of the wealth tax was followed by a break in the measurement of wealth in the administrative data. Since I do not look at wealth outcomes but rather labor market choices at the top of the distribution of wealth. I am able to identify effect of the wealth tax abolishment on labor supply decisions.

## 3.2 Danish Administrative Data

**Why is the Danish data of interest?**

The data is useful for multiple purposes. Individual income and wealth are observed over a long period of time. Population-level coverage implies gains from observing responses even at the top of the distribution of wealth. Another key positive in favor of the data is the presence of convincing instruments due to the wealth tax reform in 1988 (Jakobsen et al. 2020). Evidence from the joint distribution of income, wealth, and consumption can shed light on competing theories in public finance.

### What is exactly needed from the data?

Karaivanov & Townsend (2014) conduct their analysis in Thai villages on entrepreneurial households, who consume, work, and invest (in their firm). Karaivanov & Townsend (2014) run their estimates under different assumptions regarding the data availability and test the alternative model specifications. In the first instance, they estimates the competing models using the joint distribution of business assets, investment, and income data. They then replicate their analysis with consumption and income data alone. Then finally, they use data on business assets, consumption, investment, and income all-together. Based on maximum likelihood and other goodness of fit criterion, they conclude which model(s) fit the data best. A similar analysis in the Danish administrative data would require data at least two of the following: income data, consumption data, and wealth data.

The additional gain from using the Danish data would be accessing the labor income and assessing how taxable income responds to changes in the wealth tax. This can then be related to potential responses from the dynamic Mirreleesian model and that of the full insurance benchmark.

The size of the wealth tax corresponds closely to estimates of the theoretically optimal tax on wealth. It is also the case that this is one of the largest wealth taxes being removed.

### Concerns

- *Old Individuals and their labor response?*

  One concern raised is that the labor response might be zero due to the presence of many retired individuals in the treated subsamples. This is one way to think of it. An alternative way is to think of the retired individuals as individuals with a certain $\theta_t = 0 \quad \forall t > t_r$ where $t_r$ is the age of retirement. For those individuals, we are certain the inverse Euler equation doesn't apply since their ability measure is public information. Hence, assuming retirement is an absorbing state (as assumed in some papers in public finance (Golosov & Tsyvinski 2006)), older individuals might be a good reference sample compared to individuals who work and hence have the option of this behavioral adjustment channel. This of course would have to assume that mortality risk is not a concern and that retirement decisions are exogenously fixed.

- *Rich individuals and their labor response?*

  I share this concern that the meaningful margin of response might not be the labor supply margin for the wealth tax reforms conducted in Denmark. One way to attempt at dealing with this would be to estimate the labor responses at different wealth levels for those affected by the reform vs. not.

  Another way to address this concern would be to use self-employment and own-business earnings as a way to approach the theoretical construct of individual ability to generate labor income. Ultimately, the theory is concerned with the effort exerted by individuals to generate income. This effort is what faces dynamic distortions if policy-makers do not tax wealth.

# 4 Empirical Strategy

## 4.1 Reduced-form Effect of Wealth Tax on Labor Supply

Regression of Interest:

$$L_{it} = \sum_{j \neq 1998} \delta_j \; . \; Year_j \; . \; Treat_{it} + \gamma_i + \eta_t + v_{it} \tag{1}$$

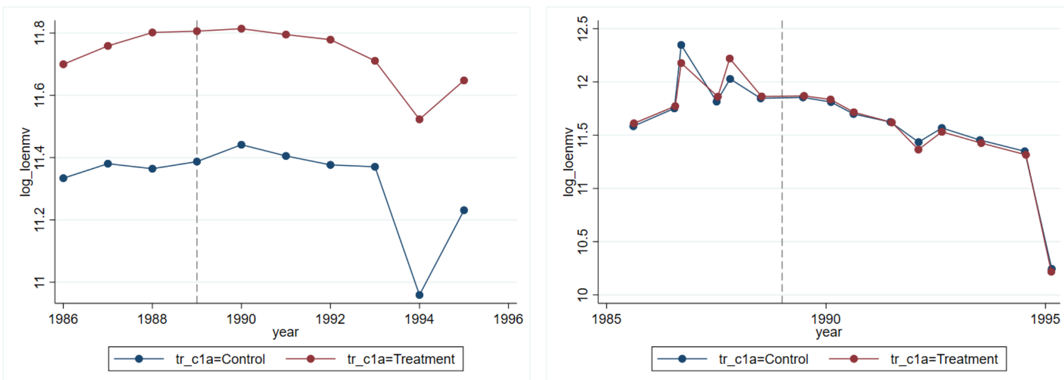
$L_{it}$ is a measure of labor supply. A list of examples are:

- Logged household taxable labor income.
- Individual and Spousal retirement decisions.

- Measure of self-employment and income from self-employment (conditonal on being self-employed).

Assuming parallel trends hold and if $\delta_{ITT} = 0$, the economic significance of the labor supply mechanism to tax wealth is weak.

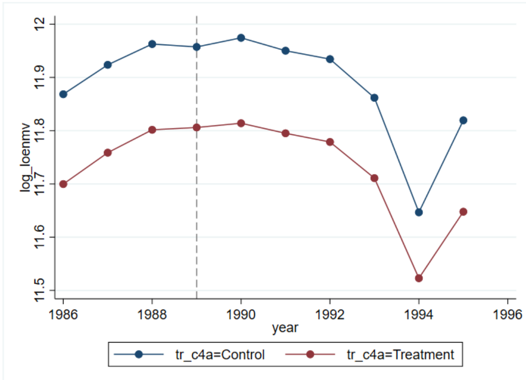
# 5 Data Exploration based on Reduced form section



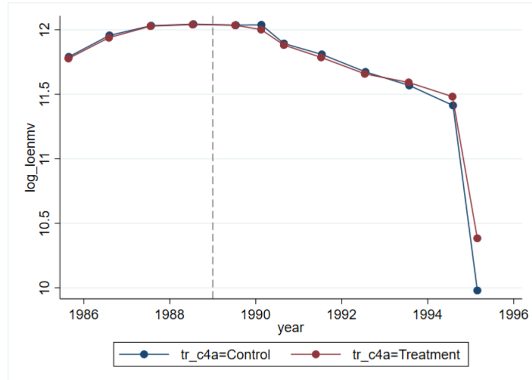

(i) Specification 1 (ii) Specification 2

Figure 1: Logged household labor income using ID1.
*Notes.* Treated households refers to married households whose marginal wealth tax rate fell from 2.2% to 0% in 1989. Control households refers to singles whose wealth tax rate continued to be 2.2% in 1989 and fell to 1% through to 1991. Specification 1 refers to a Diff-in-Diff specification without household-level fixed effects. Specification 2 adds household-level fixed effects.

This early exercise indicates that the wealth tax changes have little effect on labor incomes. Pending confirming this along multiple margins of response and using the other identification strategies, this might lead us to believe that the New Dynamic Public Finance argument to tax wealth does not really matter in practice.




(i) Specification 1




(ii) Specification 2

Figure 2: Logged household labor income using ID2.
*Notes.* Treated households refers to married households whose marginal wealth tax rate fell from 2.2% to 0% in 1989. Control households refers to married households who were below the wealth tax threshold and paid 0% wealth tax in 1989 but were part of the top 5% of wealth in 1989. Specification 1 refers to a Diff-in-Diff specification without household-level fixed effects. Specification 2 adds household-level fixed effects.

# 6 Results

# 7 Conclusion

Debates on Wealth taxation gained renewed interest. Recently, two American presidential candidates backed high wealth tax rates on the very richest Americans. In fact, opinion polls suggest that Americans would vote for a tax on the wealth of the superrich.

Leading academic arguments to tax wealth from the personal tax code usually rest on strong normative assumptions, such as taxes on inheritance (Farhi & Werning 2013*a*) and redistribution of capital income risk (Shourideh 2013).

This paper sets aside normative rationales for wealth taxation and investigates empirical evidence for a key positive argument to tax wealth in the literature. It aims to answer the question of whether the presence of private information and stochastic evolution of individual ability is corroborated in a developed economy context. If so, it aims to answer the question on the size of the labor response caused by a change in the wealth tax. The paper would

Table 1: Elasticity estimates for multiple outcome variables under different identification strategies

| | 1989 Wealth Tax Reform | | | | 1997 Wealth Tax Abolishment | |
|---|---|---|---|---|---|---|
| Elasticities: $\epsilon$ *wrt (1-$\tau_w$)* | Couples within vs. Couples Below | | Couples Within vs. Singles Below | | Tax-payers vs non-payers | |
| Dependent Variables | (1) | (2) | (3) | (4) | (5) | (6) |
| HH Labor Income | -.00217 | -.00241 | -.00301 | -.00208 | -.00233 | -.00153 |
| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
| $2^{nd}$ Earner's Labor Income | -.00217 | -.00241 | -.00301 | -.00208 | -.00233 | -.00153 |
| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
| HH Total Income | -.00217 | -.00241 | -.00301 | -.00208 | -.00233 | -.00153 |
| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
| $2^{nd}$ Earner's Total Income | -.00217 | -.00241 | -.00301 | -.00208 | -.00233 | -.00153 |
| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
| Retired (Main Earner) | -.00217 | -.00241 | -.00301 | -.00208 | -.00233 | -.00153 |
| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
| Retired ($2^{nd}$ Earner) | -.00217 | -.00241 | -.00301 | -.00208 | -.00233 | -.00153 |
| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
| Self employed (Main Earner) | -.00217 | -.00241 | -.00301 | -.00208 | -.00233 | -.00153 |
| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
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| | (.00202) | (.0021) | (.00232) | (.00523) | (.00534) | (.00482) |
| *Matching* | N | N | N | N | N | N |
| *Controls* | N | N | N | N | N | N |
| *HH Fixed Effects* | N | Y | N | Y | N | Y |

*Notes.*

contribute to the policy debate surrounding wealth taxes. It would thus influence the academic literature by assessing the need for better positive theories of capital taxation.

# Overview

**Summary of Questions:**

- Is there an empirical ground for motivating capital taxation using the Inverse Euler Equation logic in a highly-developed economy?
- Are labor supply incentives negatively affected in response to a reduction in capital taxation?

**Contributions:**

- Estimate the labor supply response to a wealth taxation in Denmark. Estimate will not suffer from within country location spillovers and, given access to population-level data, the estimate will be able to look at different margins of response.
- Test and Contrast a dynamic incomplete markets model with private information against a full insurance benchmark in a developed country context.
- Provide evidence on a key argument to tax wealth on efficiency grounds. Compare the labor supply response to that of the competing models.

# References


Adam, S. & Miller, H. (2020), The economics of a wealth tax, Technical report.

Attanasio, O. P. & Pavoni, N. (2011), 'Risk sharing in private information models with asset accumulation: Explaining the excess smoothness of consumption', *Econometrica* **79**(4), 1027–1068.
**URL:** *https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA7063*

Attanasio, O. P. & Weber, G. (2010), 'Consumption and saving: Models of intertemporal allocation and their implications for public policy', *Journal of Economic Literature* **48**(3), 693–751.
**URL:** *https://www.aeaweb.org/articles?id=10.1257/jel.48.3.693*

Blundell, R., Pistaferri, L. & Preston, I. (2008), 'Consumption inequality and partial insurance', *American Economic Review* **98**(5), 1887–1921.
**URL:** *https://www.aeaweb.org/articles?id=10.1257/aer.98.5.1887*

Brülhart, M., Gruber, J., Krapf, M. & Schmidheiny, K. (2019), Behavioral responses to wealth taxes: evidence from switzerland, Technical report, CESifo Working Paper.

Campbell, J. & Deaton, A. (1989), 'Why is consumption so smooth?', *The Review of Economic Studies* **56**(3), 357–373.
**URL:** *http://www.jstor.org/stable/2297552*

Conesa, J. C., Kitao, S. & Krueger, D. (2009), 'Taxing capital? not a bad idea after all!', *American Economic Review* **99**(1), 25–48.
**URL:** *https://www.aeaweb.org/articles?id=10.1257/aer.99.1.25*

Deaton, A. & Paxson, C. (1994), 'Intertemporal choice and inequality', *Journal of political economy* **102**(3), 437–467.

Farhi, E. & Werning, I. (2012), 'Capital taxation: Quantitative explorations of the inverse euler equation', *Journal of Political Economy* **120**(3), 398–445.
**URL:** *http://www.jstor.org/stable/10.1086/666747*

Farhi, E. & Werning, I. (2013*a*), 'Estate taxation with altruism heterogeneity', *American Economic Review* **103**(3), 489–95.
**URL:** *https://www.aeaweb.org/articles?id=10.1257/aer.103.3.489*

Farhi, E. & Werning, I. (2013*b*), 'Insurance and taxation over the life cycle', *The Review of Economic Studies* **80**(2 (283)), 596–635.
**URL:** *http://www.jstor.org/stable/43551497*

Golosov, M., Kocherlakota, N. & Tsyvinski, A. (2003), 'Optimal indirect and capital taxation', *The Review of Economic Studies* **70**(3), 569–587.
**URL:** *http://www.jstor.org/stable/3648601*

Golosov, M. & Tsyvinski, A. (2006), 'Designing optimal disability insurance: A case for asset testing', *Journal of Political Economy* **114**(2), 257–279.
**URL:** *http://www.jstor.org/stable/10.1086/500549*

Golosov, M. & Tsyvinski, A. (2015), 'Policy implications of dynamic public finance', *Annual Review of Economics* **7**(1), 147–171.
**URL:** *https://doi.org/10.1146/annurev-economics-080614-115538*

Golosov, M., Tsyvinski, A., Werning, I., Diamond, P. & Judd, K. L. (2006), 'New dynamic public finance: A user's guide [with comments and discussion]', *NBER macroeconomics annual* **21**, 317–387.

Heathcote, J., Storesletten, K. & Violante, G. L. (2014), 'Consumption and labor supply with partial insurance: An analytical framework', *American Economic Review* **104**(7), 2075–2126.

Jakobsen, K., Jakobsen, K., Kleven, H. & Zucman, G. (2020), 'Wealth Taxation and Wealth Accumulation: Theory and Evidence From Denmark*', *The Quarterly Journal of Economics* **135**(1), 329–388.
**URL:** *https://doi.org/10.1093/qje/qjz032*

Karaivanov, A. & Townsend, R. M. (2014), 'Dynamic financial constraints: Distinguishing mechanism design from exogenously incomplete regimes', *Econometrica* **82**(3), 887–959.
**URL:** *https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA9126*

Keane, M. P. (2011), 'Labor supply and taxes: A survey', *Journal of Economic Literature* **49**(4), 961–1075.
**URL:** *https://www.aeaweb.org/articles?id=10.1257/jel.49.4.961*

Kinnan, C. (2021), 'Distinguishing barriers to insurance in thai villages', *Journal of Human Resources* pp. 0219–10067R1.

Ligon, E. (1998), 'Risk sharing and information in village economies', *Review of Economic Studies* **65**(4), 847–864.
**URL:** *https://EconPapers.repec.org/RePEc:oup:restud:v:65:y:1998:i:4:p:847-864.*

Mirrlees, J. A. (1971), 'An exploration in the theory of optimum income taxation', *The Review of Economic Studies* **38**(2), 175–208.
**URL:** *http://www.jstor.org/stable/2296779*

Ring, M. A. K. (2020), 'Wealth taxation and household saving: Evidence from assessment discontinuities in norway', *Available at SSRN 3716257* .

Rogerson, W. P. (1985), 'Repeated moral hazard', *Econometrica* **53**(1), 69–76.
**URL:** *http://www.jstor.org/stable/1911724*

Saez, E. (2001), 'Using Elasticities to Derive Optimal Income Tax Rates', *The Review of Economic Studies* **68**(1), 205–229.
**URL:** *https://doi.org/10.1111/1467-937X.00166*

Saez, E. & Zucman, G. (2020), 'The rise of income and wealth inequality in america: Evidence from distributional macroeconomic accounts', *Journal of Economic Perspectives* **34**(4), 3–26.

Scheuer, F. & Slemrod, J. (2021), 'Taxing our wealth', *Journal of Economic Perspectives* **35**(1), 207–30.
**URL:** *https://www.aeaweb.org/articles?id=10.1257/jep.35.1.207*

Seim, D. (2017), 'Behavioral responses to wealth taxes: Evidence from sweden', *American Economic Journal: Economic Policy* **9**(4), 395–421.
**URL:** *https://www.aeaweb.org/articles?id=10.1257/pol.20150290*

Shourideh, A. (2013), Optimal Taxation of Wealthy Individuals, Technical report.

Townsend, R. M. (1994), 'Risk and insurance in village india', *Econometrica* **62**(3), 539–591.
**URL:** *http://www.jstor.org/stable/2951659*

# A Overview of Efficiency Grounds to Taxing Wealth under New Dynamic Public Finance

An important argument in the literature to tax wealth relies on an informational inefficiency that diminishes the incentive of individuals to supply labor at the optimal level. Taxing wealth hence corrects incentives to supply labor. In this "new dynamic public finance" argument, the presence of a moral hazard informational inefficiency, due to the stochastic evolution of an agent's ability and its lack of observability by a planner of an individual's ability to generate labor income, leads to an endogenously incomplete market. In this world, the optimal decision for an individual is to make consumption and labor supply plans according to the inverse Euler equation , as opposed to the standard Euler Equation (which solves the optimal allocation problem of the planner). The two Euler equations are listed as follows,

$$E\{\beta(1+r_{t+1})U'(C_{t+1}(\theta_{t+1}))\} = U'(C_t(\theta_t)) \qquad \text{Planner's Euler Equation}$$

$$\frac{1}{\beta(1+r_{t+1})}\mathrm{E}_t\left[\frac{1}{U'(C_{t+1}(\theta_{t+1}))}\right] = \frac{1}{U'(C_t(\theta_t))} \qquad \text{Inverse Euler Equation}$$

By Jensen's inequality, the inverse Euler Equation implies the need for a intertemporal wedge (tax) on wealth. Intuitively, as opposed to a full information benchmark, left to their own devices, individuals would save too much relative to the constrained pareto optimal allocation. Alternatively, the desired labor supply at the optimal allocation is incompatible with free savings. This is because an increase in wealth has two effects on welfare in the following period. Although it increases the resources available for the individual in the second period, it has negative impact on labor supply incentives. This paper provides evidence on whether this argument to tax wealth is relevant for policy design.

Suppose we live in an economy with private information, stochastic ability, and incomplete markets, what should this wealth tax be to achieve the constrained pareto optima? Multiple papers in the macro-public finance tradition attempt to provide an answer for this question using quantitative exercises in a partial or general equilibrium framework. Under standard parametric assumptions, if consumption log-normally distributed with a variance $\sigma_\epsilon^2$, Farhi & Werning (2012) characterise the optimal capital distortion as

$$\tau_t^K = 1 - exp(-\sigma_\epsilon^2) \qquad \text{Optimal Wealth Distortions}$$

so that $\tau_t^K \approx \sigma_\epsilon^2$, when $\sigma_\epsilon$ is small. Empirical estimates, using the panel study of income dynamics, of the size of $\sigma_\epsilon^2$ suggest values between 0.56% and 1% (Deaton & Paxson 1994, Blundell et al. 2008, Heathcote et al. 2014). Conesa et al. (2009) in another quantitative exercise, using a dynamic general equilibrium model, estimate the optimal capital income tax to be around 36%, which translates to 2% if the tax was implemented on capital as opposed to capital income. These estimates are close to the magnitude of the wealth tax change studied in this paper.

# B Ligon (1998) Exercise: Distinguishing Permanent Income and Private Insurance

This appendix details work-in-progress assessing directly which of the competing models of household behavior apply – a finding in support of the self-insurance or complete markets would be consistent with the reduced form evidence in the paper and find limited support for taxing wealth on efficiency grounds related to the supply of labor. The exercise follows Ligon (1998). Here, I distinguish whether a self-insurance model explains the data better or a private information model. The test would fail to provide implications under full insurance. Assuming $\beta * R = 1$ and $U(c) = c^{1-\gamma}/(1-\gamma)$, where c is consumption, the difference between a self-insurance model and a model with private information can be described by estimating $b_0$ in the following equation, using GMM:

$$\mathrm{E}_t\left[\left(\frac{c_{i,t+1}}{c_{i,t}}\right)^{b_0} - 1\right] = 0 \qquad \text{(Test of the Euler Equation)}$$

Suppose we observe a sample $c_{it}$ for $t = 1, ..., T-1$ years and for $i = 1, ..., N$ households. Separately on a pre-reform treated sample, post-reform treated sample, pre-reform control sample, and a post-reform control sample, do the following:

- Estimate $b_0$ by the Generalised Method of Moments (GMM).

$$\mathrm{E}\left[\left(\frac{c_{i,t+1}}{c_{i,t}}\right)^{b_0} - 1\right] = 0 \tag{2}$$

- Next period's consumption is endogenous, instrument using $Z_{it} \in$ past income, past wealth.

- Estimate $b_0$ and $J$-statistic by GMM. Under rational expectations, the identifying restrictions are given by:

$$\mathrm{E}\left[\left(\left(\frac{c_{i,t+1}}{c_{i,t}}\right)^{b_0} - 1\right).Z_{it}\right] = 0 \tag{3}$$

where the equation of interest is:

$$\mathrm{E}\left[\left(\frac{c_{i,t+1}}{c_{i,t}}\right)^{b_0} - 1\right] = v_{t+1} \tag{4}$$

- If $b_0 < 0$ at the relevant significance level. Then infer that data was generated by permanent income hypothesis.
- If $b_0 > 0$ at the relevant significance level. Then infer that data was generated by private information economy.